

Extension of scalars

Given a finite field extension K of $\mathbb{Q}$ and a $\mathbb{Q}$ -basis $S = \{\alpha_1, \dots, \alpha_n\}$ of K , we have defined a ring inclusion $M_S : K \rightarrow M_n(\mathbb{Q})$ where the latter is the ring of $n \times n$ matrices with entries in $\mathbb{Q}$. Here $M_S(\alpha) = (a_{i,j})$ where

$$\alpha \cdot \alpha_i = \sum_{j=1}^n a_{i,j} \alpha_j$$

Alternatively, we write

$$\alpha_p \cdot \alpha_q = \sum_{r=1}^n c_{p,q,r} \alpha_r$$

where $c_{p,q,r} \in \mathbb{Q}$ are sometimes called the *structure constants* of K as a $\mathbb{Q}$ -algebra. Note that $M_S(\alpha_p) = (c_{p,i,j})$ so if $\alpha = \sum_{p=1}^n a_p \alpha_p$, then $M_S(\alpha) = (a_{i,j})$ where

$$a_{i,j} = \sum_{p=1}^n a_p c_{p,i,j}$$

Hence M_S is determined by the structure constants. Thus, K is determined as a $\mathbb{Q}$ -algebra by the structure constants once we are given a basis of K as a $\mathbb{Q}$ -vector space.

Given $\mathbb{Q}$ as a subfield of a field E , we can consider $c_{p,q,r}$ as elements of E . This defines an E -algebra structure on $E \otimes_{\mathbb{Q}} K = E\alpha_1 + \dots + E\alpha_n$. Equivalently, we can define $E \otimes_{\mathbb{Q}} K$ as the subset $E \cdot M_S(K)$ of $M_n(E)$.

Exercise: Check that a different choice of basis gives an isomorphic algebra.

Thus, $E \otimes_{\mathbb{Q}} K$ is well-defined independent of the choice of basis.

Algebraic integers

If $\mathcal{O}_K$ denotes the ring of algebraic integers in K , then we have seen that $S = \{\alpha_1, \dots, \alpha_n\}$ can be chosen so that $\mathcal{O}_K = \mathbb{Z}\alpha_1 + \dots + \mathbb{Z}\alpha_n$. This means that $c_{p,q,r}$ are in $\mathbb{Z}$ and describe $\mathcal{O}_K$ as a $\mathbb{Z}$ -algebra (which is just a ring, since any (unital) ring is a $\mathbb{Z}$ -algebra!).

The image of $\mathcal{O}_K$ via M_S is in $M_n(\mathbb{Z})$; in fact, we have $M_S(\mathcal{O}_K) = M_S(K) \cap M_n(\mathbb{Z})$.

It follows that $E \cdot M_S(K) = E \cdot M_S(\mathcal{O}_K)$. Hence we can identify $E \otimes_{\mathbb{Q}} K$ with $E \otimes_{\mathbb{Z}} \mathcal{O}_K$.

Decomposition

The E -algebra $A = E \otimes_{\mathbb{Q}} K$ is a finite dimensional E -vector space. By the structure theorem about such algebras, it has finitely many maximal ideals

$M_1, \dots, M_k$ such that we have an isomorphism:

$$A \cong \frac{A}{M_1^{n_1}} \times \dots \times \frac{A}{M_k^{n_k}}$$

where n_i are some positive integers.

We have the trace map $\text{Tr}_{K/\mathbb{Q}} : K \rightarrow \mathbb{Q}$ given by $\text{Tr}_{K/\mathbb{Q}}(\alpha) = \text{Trace} M_S(\alpha)$. This extends to the trace map $\text{Tr}_{A/E} : A \rightarrow E$ by linearity. Similarly, the trace pairing $(\alpha, \beta) = \text{Tr}_{K/\mathbb{Q}}(\alpha \cdot \beta)$ extends to a pairing $A \times A \rightarrow E$ which is given by the same symmetric invertible matrix $B = (b_{i,j})$ where $b_{i,j} = (\alpha_i, \alpha_j)$. It follows that, the pairing is non-degenerate.

Now if $\eta \in A$ were to be nilpotent, then $\alpha\eta$ would be nilpotent for all $\alpha \in A$ (since A is commutative). This would mean that $(\alpha, \eta) = \text{Trace} M_S(\alpha\eta) = 0$ all $\alpha \in A$; which contradicts the non-degeneracy of the pairing.

No element in A is nilpotent.

As a consequence, we see that

$$A \cong \frac{A}{M_1} \times \dots \times \frac{A}{M_k}$$

is a product of fields $E_i = A/M_i$ which are finite extensions of E of degree n_i where $\sum_{i=1}^k n_i = n$.

Case E is $\mathbb{C}$ or $\mathbb{R}$

In particular, if $E = \mathbb{C}$, then $A = \mathbb{C} \times \dots \times \mathbb{C}$ is a product of n copies of $\mathbb{C}$ as a $\mathbb{C}$ -algebra. The same applies to any algebraically closed field containing $\mathbb{Q}$. In terms of M_S , this isomorphism can be seen as the change of basis which *simultaneously* diagonalises all $M_S(\alpha_i)$ for $i = 1, \dots, n$; equivalently, we have a matrix $P \in M_n(\mathbb{C})$ such that $P \cdot M_S(\alpha) \cdot P^{-1}$ is a diagonal matrix in $M_n(\mathbb{C})$ for all $\alpha \in K$. Note that the diagonal entries are the roots of the characteristic polynomial $\text{ch}_{K/\mathbb{Q}}(\alpha)$ of $M_S(\alpha)$.

When $E = \mathbb{R}$, then the only finite extensions of E are $\mathbb{R}$ and $\mathbb{C}$ (upto isomorphism). Thus, there are non-negative integers r_1 and r_2 such that $r_1 + 2r_2 = n$ and

$$\mathbb{R} \otimes_{\mathbb{Q}} K \cong \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$$

as $\mathbb{R}$ -algebras.

Case E is $\mathbb{Q}_p$

We now examine $A = \mathbb{Q}_p \otimes_{\mathbb{Z}} \mathcal{O}_K$. As seen above $A = L_1 \times \dots \times L_g$, where each L_i is a finite extension of $\mathbb{Q}_p$, say of degree n_i ; we then have $n = \sum_{i=1}^g n_i$.

In a separate section of the notes for this course we have seen the following “structure results” for L_i :

- $|\alpha|_p = |\text{Nm}_{L_i/\mathbb{Q}_p}(\alpha)|_p^{1/n_i}$ defines an ultra-metric absolute value on L_i that extends the usual absolute value on $\mathbb{Q}_p$. This metric is equivalent to the $\mathbb{Q}_p$ -vector space metric on L_i .
- The ring $R_i = \{\alpha : |\alpha|_p \leq 1\}$ is the collection of all elements of L_i which are integral over $\mathbb{Z}_p$.
- We have an element u_i in R_i such that

$$R_i = \mathbb{Z}_p[u_i] \cong \frac{\mathbb{Z}_p[X]}{\langle f_i(X) \rangle}$$

where $f_i(X)$ is the minimal polynomial of u_i ; this polynomial is monic of degree n_i .

- There is a monic polynomial $g_i(X) \in \mathbb{Z}_p[X]$ of degree f_i , so that:
 - $g_i(X) \bmod p$ is irreducible and
 - $f_i(X) \bmod p = g_i(X)^{e_i} \bmod p$ for some positive integer e_i .

(Note that this means $n_i = e_i \cdot f_i$.)

- There is a unique non-zero prime ideal in R_i which is generated by $\pi_i = g_i(u_i)$. Every element of L_i is uniquely of the form $v \cdot \pi_i^k$ where $v \in R_i$ is a unit and k is an integer. Moreover, p is a unit multiple of $\pi_i^{e_i}$.

Now $\mathbb{Q}$ is dense in $\mathbb{Q}_p$ so it follows that K is dense in $A = \mathbb{Q}_p \otimes_{\mathbb{Q}} K$ when it is considered as a vector space over $\mathbb{Q}_p$. The decomposition $A = L_1 \times \cdots \times L_g$ is compatible with this topology in the sense that the metric on the right is equivalent to the vector space metric.

The product U of the open sets $U_i = u_i + \pi_i^2 R_i$ in L_i is a non-empty open set in A . Thus it contains an element $\eta \in K$. The image v_i of η in L_i can be seen as an “alternative” to u_i in the sense that:

- v_i is an element of R_i whose minimal polynomial $h_i(X)$ satisfies $h_i(X) \bmod p = g_i(X) \bmod p$.
- $\tau_i = g_i(v_i)$ is a unit multiple of π_i .
- $R_i = \mathbb{Z}_p[v_i]$.

If $h(X) \in \mathbb{Q}[X]$ is the characteristic polynomial of η , then we see that $h(X) = h_1(X) \cdots h_g(X)$ in $\mathbb{Q}_p[X]$; where the left-hand side is in $\mathbb{Q}[X]$ and the right-hand side is in $\mathbb{Z}_p[X]$. It follows that $h(X)$ lies in $\mathbb{Q} \cap \mathbb{Z}_p = \mathbb{Z}_{(p)}$ which consists of rational numbers whose denominators are co-prime to p . In other words, η is integral over $\mathbb{Z}_{(p)}$.

Replacing η by an integer multiple co-prime to p , we can assume that η is integral over $\mathbb{Z}$; i.e. it is in $\mathcal{O}_K$.

In summary, there is an element η in $\mathcal{O}_K$ such that, for each $i = 1, \dots, g$, its image v_i in L_i satisfies $R_i = \mathbb{Z}_p[v_i]$. Since the image of $\mathcal{O}_K$ in L_i is clearly

integral over $\mathbb{Z}_p$, we see that

$$\mathbb{Z}_p \otimes_{\mathbb{Z}} \mathcal{O}_K = R_1 \times \cdots \times R_g \subset L_1 \times \cdots \times L_g = A$$

We denote the left-hand expression as $\mathcal{O}_A$.

For each i , we can take a lift $g_i(X)$ of $g_i(X) \bmod p$ so that $g_i(X) \in \mathbb{Z}[X]$. Thus $\pi_i = g_i(\eta)$ is an element of $\mathcal{O}_K$ whose image in L_i (which we denote by the same symbol π_i !) has the above properties.

For later use, we note that we have a elements η and $\pi_1, \dots, \pi_g$ of $\mathcal{O}_K$ such that:

- The image v_i of η in L_i generates R_i over $\mathbb{Z}_p$.
- The image of π_i in L_i is such that every element of L_i is of the form $w \cdot \pi_i^k$ for some unit w in R_i and some integer k .